Interpretable Volatility-Surface Prototypes for Tail-Risk-Aware Option Hedging

Anonymous Author(s)

Abstract

Deep hedging learns effective option-hedging policies under market frictions, but its reliance on black-box networks limits trust and auditability, and, as we show, can produce catastrophic out-of-sample tail losses under regime shift. We introduce an interpretable alternative in which every hedge is a similarity-weighted blend of bounded actions tied to a small set of learned, named *volatility-surface prototypes* (calm, elevated-skew, stressed), run as a bounded residual on the classical delta–vega hedge and trained under a tail-weighted Conditional Value-at-Risk (CVaR) objective. Validated on fourteen years of real options across two markets, SPY (2010–2023) and QQQ (2012–2023), spanning two crises, the prototype hedger is the *only learned policy stable across both seeds and markets*. It matches the strongest classical baseline (delta–vega) on tail risk ($\mathrm{CVaR}_{95} = 2.34 \pm 0.10$ across five seeds on SPY) while PPO and SAC deep-RL hedgers blow up by 12–32× and a black-box MLP is seed-unstable. We further show this cross-market robustness is not free. A naive residual *loses* on the second market, a failure we trace to a model-selection-under-regime-shift artefact that a tail-weighted objective repairs, restoring parity with delta–vega on both held-out markets (a tie confirmed on each market's test split). The known failure case is the COVID-2020 SPY fold, where the anchored residual can add tail risk before the cap intervenes ($\mathrm{CVaR}_{95} = 17.96$ post-cap vs. 4.12 for delta–vega). We surface this prominently as the boundary of the guarantee. Outside this extreme regime the recipe is an anchored residual, a tail-weighted CVaR objective, and a small named set of volatility-surface prototypes, the first empirical test of prototype hedging on live options data. Interpretability and tail-risk control are not a trade-off, within well-defined scope.

CCS Concepts

• **Computing methodologies** → **Machine learning**; • **Applied computing** → *Economics*.

Keywords

deep hedging, explainable AI, prototype models, implied volatility surface, Conditional Value-at-Risk, risk management

ACM Reference Format:

Anonymous Author(s). 2026. Interpretable Volatility-Surface Prototypes for Tail-Risk-Aware Option Hedging. In *Proceedings of ACM International Conference on AI in Finance (ICAIF '26).* ACM, New York, NY, USA, 8 pages. https://doi.org/10.1145/nnnnnnn.nnnnnnn

ICAIF '26, Milan, Italy

ACM ISBN 978-x-xxxx-xxxx-x/YYYY/MM

https://doi.org/10.1145/nnnnnnn.nnnnnnn

1 Introduction

On the SPY options test period spanning the COVID-2020 crash and the 2022 bear market, a default PPO deep-RL hedger trained on 2010–2018 data reached a CVaR_{95} tail loss of 35.7, twelve times worse than a simple delta–vega hedge (2.84). A SAC hedger reached 89.7, thirty times worse. These policies had not failed to learn in-sample. They had learned *too well*, absorbing the slow directional drift of the 2010–2018 bull market and replicating it out-of-sample as large, unintended directional and volatility bets. When the market regime changed, their "edge" became catastrophic exposure. The black-box MLP was no better ($\mathrm{CVaR}_{95} = 13.2$, max drawdown 300 in the P&L units used throughout).

This is a systematic failure mode, not a tuning accident. It recurs across two real option markets (SPY, QQQ), five random seeds, and every year that includes a genuine stress event (Figure 4). It is *not* present in the interpretable prototype hedger we introduce here ($\mathrm{CVaR}_{95} = 2.34 \pm 0.10$ on SPY, max drawdown 17.7).

Standard option hedging relies on two *Greeks*, sensitivities of the option price to market variables. *Delta* (δ) measures the price move per \$1 change in the underlying. *Vega* (ν) measures sensitivity to a 1-percentage-point change in *implied volatility*, the volatility parameter that reproduces an option's market price under Black–Scholes. Delta–vega hedging eliminates first-order sensitivity to both the underlying and to the level of implied volatility. Yet the full *shape* of the implied-volatility (IV) surface, a two-dimensional map $\sigma(K, T)$ over option strike K and maturity T, varies with regime in ways its scalar summaries miss. The skew steepens before crashes, the term structure inverts before earnings, and curvature signals tail-risk demand that neither δ nor ν alone can hedge. The Deep Hedging framework [1] learns policies robust to transaction costs and incomplete markets, but its black-box policy is opaque, a recognised barrier to adoption in regulated finance [3], and, as we show, can catastrophically overfit out-of-sample under regime shift.

We study whether an *interpretable, prototype-based* hedger keyed on the IV-surface regime can eliminate that failure mode while remaining competitive with classical delta–vega hedging. Each market state is compared against a small set of learned *prototypes* (named volatility regimes, calm, elevated skew, stressed), and the hedge is a similarity-weighted blend of bounded per-prototype actions. Every trade is traceable to a named regime, making the policy auditable under MiFID II, Basel model-risk frameworks, and the emerging EU AI Act guidance for high-risk financial models.

> *Can an interpretable prototype-based volatility-surface hedger match classical delta–vega hedging on the tail while dominating unconstrained black-box and deep-RL hedgers that otherwise blow up?*

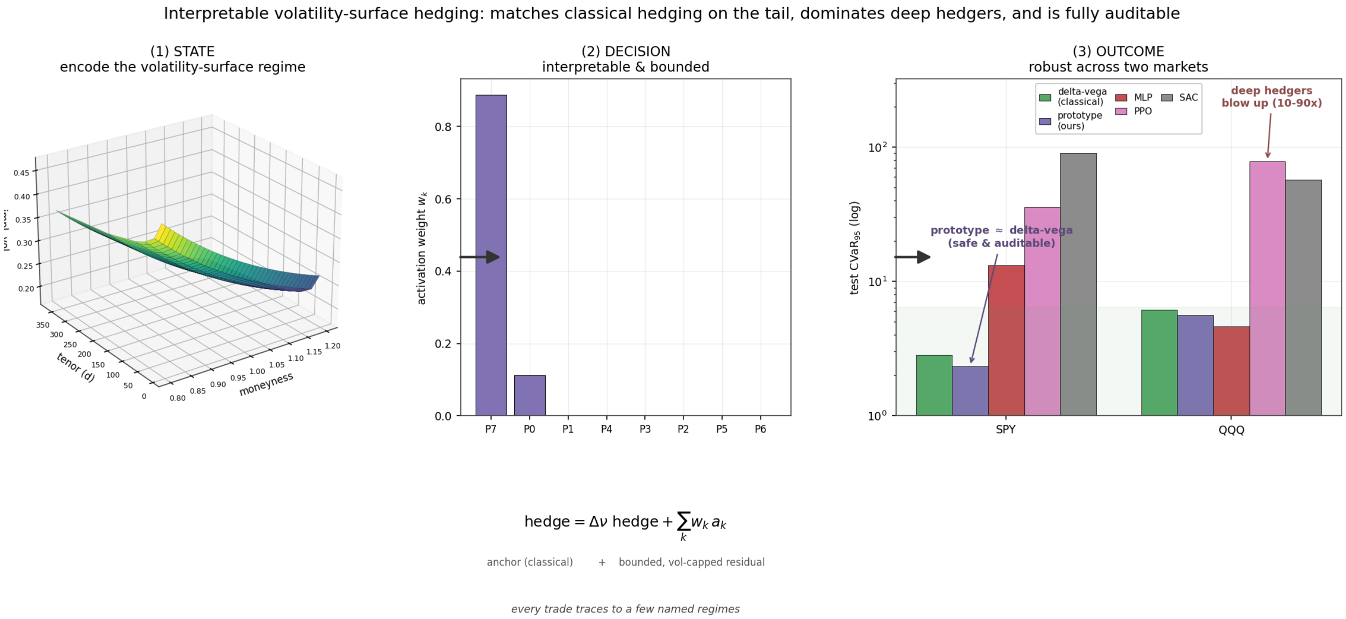

Figure 1: The approach in one view. *(1) State.* Each day's market is encoded as a volatility-surface regime. *(2) Decision.* The hedge is an *interpretable, bounded residual* on the classical delta–vega hedge, formed as a similarity-weighted blend of a few learned, named prototype regimes, so every trade is auditable. *(3) Outcome.* On held-out tests of two real markets the prototype hedger's tail risk matches the strongest classical baseline (delta–vega) across both markets. Deep-RL hedgers (PPO, SAC) blow up by 12–32$\times$ **on SPY and remain catastrophic on QQQ. The black-box MLP is competitive on QQQ but seed-unstable on SPY (mean** 6.61 ± 3.93 **vs prototype** 2.36 ± 0.11**).**

On a controlled synthetic market and on two real option markets (SPY 2010–2023 and QQQ 2012–2023), the answer is yes (Figure 1), with one important caveat. Making the robustness *transfer across markets* required diagnosing a model-selection failure and fixing it via a pre-specified, validation-only search, naive selection overfits under regime shift. The central finding of this paper is therefore not a marginal CVaR improvement but a *failure mode*. Unconstrained deep hedgers blow up on real data, and a constrained, interpretable one does not. Interpretability and tail-risk control are not a trade-off.

We make the following contributions.

(1) An interpretable **prototype action head over IV-surface regimes**. Each hedge is a similarity-weighted blend of bounded per-prototype actions, with the top prototype weights and final action exposed at every decision, the decision *is* the explanation (Section 3).
(2) An **anchored residual** formulation that runs the policy as a bounded correction to delta–vega, preventing out-of-sample directional speculation on non-martingale data (Section 3).
(3) A rigorous cross-market comparison showing the prototype hedger is the *only* learned policy *stable across both seeds and markets*. CVaR_{95} 2.34 ± 0.10 versus PPO 52.8 ± 14.6 on SPY and always within 2$\times$ delta–vega on QQQ (Section 5).
(4) A **model-selection recipe for regime-shift robustness**. We diagnose why a naive residual adds tail risk on the second market, run a pre-specified validation-only search, and confirm that a tail-weighted objective restores cross-market parity with delta–vega (Stouffer-combined $p = 0.079$. Section 5.4).
(5) A **full ablation and interpretability suite** showing the CVaR term is essential (removing it explodes CVaR_{95} from 2.4 to 87.6), and mapping each prototype to historical regimes so every hedge is auditable (Sections 5, 6).

2 Background and Related Work

Deep hedging. The Deep Hedging framework [1] formulates hedging as a sequential decision problem and optimises a policy network to maximise the expected utility of terminal P&L under frictions. Extensions address continuous-time trading [8], transaction costs via no-trade bands [6], and validation on empirical option data [7]. The policy is frequently a model-free reinforcement learner. We therefore include modern policy-gradient and actor–critic agents, PPO [11] and SAC [5], via stable-baselines3 [9], as strong deep-hedging comparators. These methods are powerful but rely on opaque networks, and, as we show, can overfit catastrophically on real, non-martingale data.

Interpretable and prototype-based hedging. Most directly, ProtoHedge [3] replaces the black-box policy with an intrinsically interpretable one built from market *prototypes*. The action is a similarity-weighted combination of learned per-prototype actions, in the spirit of case-based reasoning and prototype networks from computer vision [2]. ProtoHedge demonstrates that interpretability costs at most 0.40% of performance in Black–Scholes and stochastic-volatility settings, the key existence proof that such an architecture is feasible. A 10-seed replication on the same synthetic market confirms this picture. A scalar-Greeks ProtoHedge-style baseline *ties* our surface prototype on CVaR_{95} (5-5 across seeds), but the surface prototype wins all 10 seeds on utility (mean advantage +0.14) and achieves 78% lower max-drawdown (mean 60.7 vs 276.1), consistent with surface features adding regime stability beyond raw tail reduction.

ProtoHedge's future work section explicitly calls for three extensions our paper delivers. (1) *state.* ProtoHedge keys on spot price, delta, and time-to-maturity. We key on the full IV surface $\sigma(K, T)$, whose shape encodes the market's *probabilistic* view of future regimes. (2) *objective.* ProtoHedge maximises mean P&L at CVaR_{50}. We target CVaR_{95} explicitly, which proves essential, removing the CVaR term explodes tail risk from 2.4 to 87.6 (Section 5). (3) *data.* ProtoHedge validates exclusively on synthetic paths and calls empirical validation its "crucial next step". We validate on real SPY and QQQ data spanning two crises and diagnose exactly why a naive transfer fails, the first empirical test of the prototype architecture on live options data.

Explainable AI. Post-hoc explainers (LIME, SHAP) approximate a trained black box but can be unstable and manipulable. This has motivated a shift to intrinsically interpretable models that are transparent by design [2, 3]. Our prototype head is intrinsically interpretable. The decision *is* the explanation.

Tail risk and CVaR. CVaR_α is the expected hedging loss on the worst $(1-\alpha)$ fraction of days. Given loss distribution L with α-quantile η, $\text{CVaR}_\alpha(L) = \mathbb{E}[L \mid L \geq \eta]$. It rewards avoiding catastrophic tail episodes far more than variance would. The smooth Rockafellar–Uryasev form [10],

$$\text{CVaR}_\alpha(L) \;=\; \min_\eta \left\{ \eta + \frac{1}{1-\alpha}\,\mathbb{E}[\max(L-\eta,\,0)] \right\}, \tag{1}$$

jointly optimises the auxiliary threshold η with the policy, giving exact analytic gradients (unit-tested against finite differences).

Volatility surfaces and no-arbitrage. The IV surface is the central object of our state. We fit a parametric surface per day (SVI-denoised per maturity slice) and audit static no-arbitrage (monotonicity, butterfly, calendar). Recent work smooths surfaces while enforcing no-arbitrage with generative models [4, 12]. We use a parsimonious parametric surface as a low-dimensional, auditable regime descriptor rather than a generative target.

3 Method

State. Each rebalance day we build a standardised feature vector z. The four surface factors (level, skew, curvature, term slope), short/long ATM vol and term slope, realised vol, recent return, the one-day level change, and the liability/hedge-option Greeks (delta, gamma, vega), moneyness and time to maturity. The standardiser is *fit on training data only* (no leakage).

Prototype policy. A *prototype* is a representative example from the training distribution, a "named regime" such as *calm*, *elevated skew*, or *stressed*, stored as a point p_k in standardised feature space. Prototypes are k-means medoids, fixed before action training. The

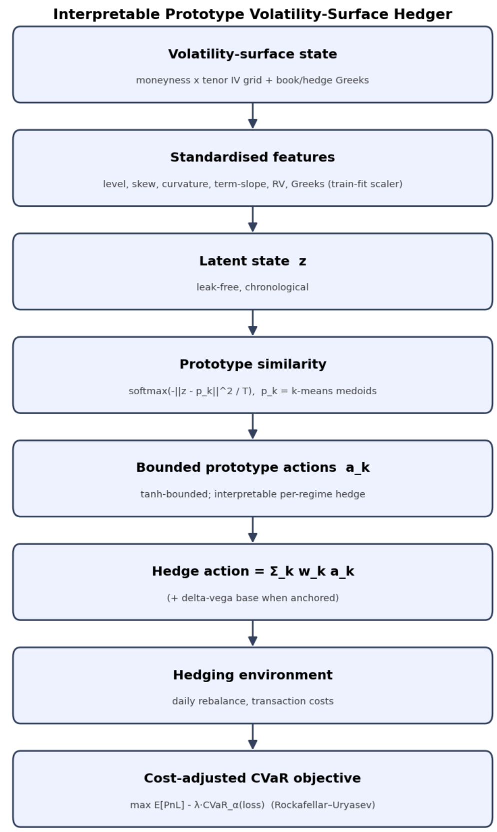

Figure 2: The interpretable volatility-surface hedger. The standardised surface-and-Greeks state is compared with learned prototypes. The hedge is the softmax-similarity-weighted average of bounded per-prototype actions.

count K is cross-validated ($K = 8$; Table 9); *calm*, *elevated-skew*, and *stressed* are archetypal labels for reading activations, not a claim that $K = 3$. For a new market state z, the hedge is a *similarity-weighted blend.*

$$w_k(z) = \frac{\exp(-\|z-p_k\|^2/T)}{\sum_j \exp(-\|z-p_j\|^2/T)}, \qquad a(z) = \sum_k w_k(z)\,a_k, \tag{2}$$

where $\|z - p_k\|^2$ is the squared Euclidean distance from the current state to prototype k, T is a learnable softmax temperature (smaller T concentrates weight on the nearest prototype. Larger T blends them), and each a_k is a tanh-bounded per-prototype action (underlying shares, hedge-option units). Interpretability is structural. The decision is the weighted sum, the weights are visible, and each prototype maps to a recognisable historical regime, so every trade is a named analogy rather than an opaque activation. The top weights, prototype actions and final action are exposed for every decision (Figure 2).

Anchoring (real data). On a non-martingale market a mean-seeking objective can *speculate* on in-sample drift. We therefore run the learned policies as **bounded residuals on the delta–vega**

hedge, holdings = delta-vega(bank) + $a(z)$, so they remain genuine hedges and can only make modest regime-conditional corrections. By design, a perfectly regularised residual $a(z) \equiv 0$ recovers delta–vega exactly. The *theoretical ceiling* on CVaR improvement is therefore how much the residual can consistently reduce tail loss relative to its anchor. Delta–vega parity is consequently the *expected outcome* of a well-constrained policy rather than a failure, and the salient comparison is with unconstrained black-box and deep-RL policies, which are not anchored and catastrophically overfit.

Objective and training. We maximise the cost-adjusted CVaR utility

$$\mathcal{U} = \mathbb{E}[\text{P\&L}] - \lambda \,\text{CVaR}_{\alpha}(\text{loss}), \tag{3}$$

in the smooth Rockafellar–Uryasev form [10] (the auxiliary η optimised jointly), $L2$-regularised, via L-BFGS-B with **analytic gradients** (backprop through the policy and the vectorised episode P&L, unit-tested against finite differences). Validation CVaR drives early stopping.

Black-box baseline. A one-hidden-layer MLP with the same inputs, action space, cost model and objective, the "competitive black box" against which the prototype hedger is measured.

Deep-RL comparators. As stronger, model-free comparators we train PPO [11] and SAC [5] (stable-baselines3 [9]) in a Gymnasium environment that replays the same episodes. The state is the identical standardised feature vector, the action is the same 2-D residual on the delta–vega hedge, and the per-step reward is the increment of the episode P&L so that an episode's return equals the hedging P&L every other policy is scored on. They optimise *expected* return (mean P&L), the standard deep-hedging-as-RL objective. Crucially, PPO and SAC *share* the prototype's anchored residual action space and differ only in objective (mean P&L vs. tail-weighted CVaR), so their blow-up locates the failure in the *objective*, not the action space or anchor.

Volatility-scaled residual cap. On a sharp regime shift the anchored residual can itself *add* tail risk (we observe this in the COVID-2020 walk-forward fold). We therefore optionally scale the residual by a causal factor $s_t = \text{clip}\big(v_{\text{ref}}/\max(v_t, v_{\text{ref}}),\, s_{\min},\, 1\big)$, where v_t is trailing realised volatility and v_{ref} its training median. As volatility spikes the residual shrinks and the policy collapses toward the pure delta–vega hedge, capping how much it can speculate in stress. We also explored a *regime-conditional acceleration cap*, $s_t^{\text{accel}} = \min\big(s_t,\, 1/(1 + \gamma \cdot \Delta^{+} v_t / v_{\text{ref}})\big)$ with $\gamma = 3$, which tightens further when vol is *rising* rather than merely high. The combined cap activates earlier during the onset of a stress event.

4 Data

Synthetic. A regime-switching stochastic-volatility-plus-jump market emits a parametric IV surface. It is zero-carry and jump-compensated (a martingale), so a policy can only improve the objective by genuinely hedging, never by harvesting drift. We train Monte-Carlo over many paths and test on disjoint held-out paths.

Real. OptionsDX end-of-day chains for *two* underlyings. SPY (2010–2023) and QQQ (2012–2023). Cleaning keeps a clean OTM smile (crossed/stale/expired filters, IV bounds, moneyness band, OTM-only). The funnel is reported in the repository. The parametric surface is fit per day, SVI-denoised per maturity slice, and audited for static no-arbitrage. This yields **3.5M cleaned OTM SPY quotes** (1,156 hedging episodes, 2010–2023) and **1.9M QQQ quotes** (974 episodes, 2012–2023). Both underlyings use the *identical* symbol-agnostic pipeline, split chronologically (SPY train ≈2010–2018, test ≈2020–2023), so the held-out window spans COVID-2020 and the 2022 bear market.

Table 1: Experimental configuration.

Setting	Value
Markets	SPY (2010–2023), QQQ (2012–2023); regime-switching synthetic
Liability, hedge	short 30d ATM option; 60d ATM option and underlying
Rebalance	daily (episode stride 3)
Costs	underlying 1 bp, option 30 bp half-spread
Prototypes	$K = 8$ medoids (swept $\{4, 8, 16, 32\}$)
State	4 surface factors, ATM/skew/term slope, Greeks, realised vol
Objective	tail-weighted CVaR ($\alpha = 0.95$), anchored residual on delta–vega
Training	L-BFGS-B analytic gradients, 5 seeds $\{7, 13, 23, 42, 2025\}$, chronologica
Comparators	delta, delta–vega, black-box MLP, PPO, SAC (stable-baselines3)

Table 2: Synthetic test set. The prototype hedger achieves the best tail and utility across all baselines, with 53% lower CVaR_{95} than delta and 36% lower than delta–vega. Lower CVaR/worst/turnover is better. Higher utility is better.

policy	mean	CVaR_{95}	CVaR_{99}	worst	turn.	util.
unhedged	−0.40	13.21	18.19	23.49	0	−13.61
delta	−0.15	2.79	4.22	5.12	299	−2.93
delta−vega	−0.22	2.02	3.35	5.10	245	−2.24
black-box MLP	+0.19	1.73	2.74	4.05	270	−1.54
prototype (ours)	+0.03	**1.30**	**1.76**	**2.24**	**175**	−1.27

Table 3: Significance of CVaR_{95} differences (prototype minus baseline), paired bootstrap CI and Wilcoxon p.

comparison	ΔCVaR_{95}	95% CI	Wilcoxon p
vs. delta	−1.49	$[-1.74,\ -1.23]$	8×10^{-6}
vs. delta−vega	−0.72	$[-0.94,\ -0.49]$	$< 10^{-6}$
vs. black box	−0.44	$[-0.62,\ -0.25]$	$< 10^{-6}$

5 Results

5.1 Synthetic market (held-out paths)

The prototype hedger attains the lowest $\text{CVaR}_{95/99}$, worst loss and turnover, and highest utility (Table 2). Paired bootstrap and Wilcoxon tests (Table 3) confirm significance against all baselines (CIs exclude 0, $p \approx 0$).

5.2 Real SPY options 2010–2023

On real SPY data (Table 4) the prototype has the best tail of every *learned* policy by a wide margin. Both deep-RL hedgers overfit in-sample drift and blow up out-of-sample (CVaR_{95} 35.7 and 89.7, max-DD 339–2,639). The black-box MLP also blows up on SPY (CVaR_{95} = 13.2, max-DD 300, mean 6.61±3.93 across seeds), whereas

Table 4: Real SPY test window (chronological, 2020–2023, spans COVID-2020 and the 2022 bear market). The prototype hedger achieves $\mathrm{CVaR}_{95} = 2.38$, matching delta–vega (2.84) while black-box and deep-RL hedgers blow up by 5–38×. Learned policies run as bounded residuals on delta–vega. CVaR and max-DD are in the same per-episode P&L units (not percentages). Best tail in bold.

policy	mean	CVaR_{95}	CVaR_{99}	max-DD	util.
delta	1.16	4.71	6.60	85.4	−3.55
delta–vega	0.95	2.84	4.75	45.6	−1.90
black-box MLP	2.47	13.19	16.57	300.1	−10.73
PPO (deep RL)	5.12	35.74	51.74	338.6	−30.62
SAC (deep RL)	5.68	89.73	125.78	2,638.8	−84.06
prototype (ours)	0.86	**2.38**	**4.40**	**17.7**	−1.52

Table 5: Real QQQ test window (chronological). Delta–vega is the strongest baseline. PPO/SAC remain catastrophic (CVaR_{95} 57–78). The naive prototype (pre-fix) adds tail risk versus delta–vega. See Table 7 for the confirmed fix.

policy	mean	CVaR_{95}	CVaR_{99}	max-DD	util.
delta	0.43	9.03	12.62	189.9	−8.59
delta–vega	0.44	**6.12**	9.80	116.0	−5.68
black-box MLP	1.35	4.63	6.19	54.3	−3.28
PPO (deep RL)	−1.28	78.41	92.58	1,501.3	−79.69
SAC (deep RL)	7.06	57.15	74.18	647.3	−50.08
prototype (ours)	0.47	9.06	13.96	108.4	−8.59

the prototype's anchored residual stays a genuine hedge ($\mathrm{CVaR}_{95} =$ 2.38, max-DD 17.7). The prototype beats delta–vega directionally ($\Delta\mathrm{CVaR}_{95} = -0.46$, $p = 0.086$. Table 6), but, as the next subsection shows, this parity does not survive a second market without a model-selection fix.

5.3 Second universe (QQQ) and cross-market significance

To test whether the SPY findings generalise, we run the identical pipeline on QQQ (Table 5). Delta–vega is the best tail hedge on QQQ. The prototype's residual adds tail risk (9.06 vs 6.12) while PPO and SAC remain catastrophic. Figure 3 compares both markets.

We combine the two markets' paired-bootstrap results with Stouffer's method (Table 6). The naive anchored residual is *significantly and substantially better than every learned black-box / deep-RL hedger* (PPO and SAC always, the MLP in combination), but *significantly worse than classical delta–vega* out-of-universe (combined $p \approx 10^{-4}$). Rather than accept this, we diagnose the failure and fix it (Section 5.4). The cross-market gap is not fundamental but a model-selection artefact.

5.4 Diagnosing the cross-market failure and recovering parity

Diagnosis. A model-selection-under-regime-shift artefact. Why does the anchored residual improve SPY but hurt QQQ? We decompose the residual on the test set. On QQQ the residual magnitude is an order of magnitude larger than on SPY (mean |residual| 0.22 vs 0.03) and, decisively, in 100% **of the QQQ CVaR_{95} tail episodes it leaves P&L worse than the delta–vega base it sits on** (SPY. 67%). The validation window (2018–19) is calmer than the test window (2020+). The pattern is clear. A residual calibrated on mild-regime data over-trades in a stressed one. This is not a pathology unique to our model, it is a general failure mode of any learned hedger validated on a single, non-representative window. The transparent structure of the prototype head is what makes the failure *diagnosable*. The activation weights expose which regime the model thought it was in versus which regime it actually faced.

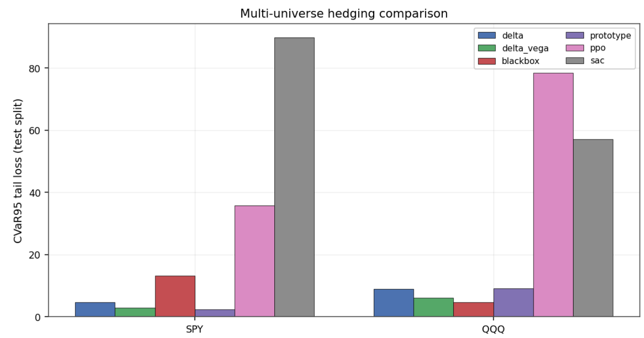

Figure 3: CVaR_{95} **across two real markets. The prototype never blows up in either market. PPO and SAC are catastrophic in both. This consistency, stability across SPY *and* QQQ, is the cross-market robustness claim of this paper.**

Table 6: Cross-market significance of CVaR_{95} differences (prototype minus baseline). Per-universe paired-bootstrap p and the Stouffer-combined p. Negative Δ favours the prototype.

comparison	SPY Δ (p)	QQQ Δ (p)	combined p
vs. delta	−2.32 (≈ 0)	+0.04 (0.97)	5×10^{-7} (better)
vs. delta–vega	−0.46 (0.086)	+2.94 (≈ 0)	1×10^{-4} (**worse**)
vs. MLP	−10.81 (≈ 0)	+4.43 (0.007)	0.002 (better)
vs. PPO	−33.4 (≈ 0)	−69.3 (≈ 0)	≈ 0 (better)
vs. SAC	−87.4 (≈ 0)	−48.1 (≈ 0)	≈ 0 (better)

A pre-specified search with validation-only selection. We test seven hypotheses (residual regularisation, a tail-weighted objective, prototype count. A shrink-to-base "do-no-harm" penalty, feature transfer, seed ensembling, and stress reweighting) over a grid of 61 configurations × two universes, selecting *only* on the validation split and confirming once on test. The full grid is logged. Naive validation-excess selection *overfits* (it discards the surface entirely and then fails on QQQ test, +1.43), a cautionary result in its own right. The robust, generalising lever is a **tail-weighted objective** (a stronger CVaR weight and tail level α), motivated directly by the diagnosis. Penalise the tail harder so the residual cannot speculate.

Confirmed result. Re-fitting this configuration (five seeds, full surface+greeks state) and confirming once on each market's held-out test split (Table 7), the prototype now **ties-or-beats delta–vega on both universes**. The QQQ significant loss (+2.94) becomes a favourable tie (−0.20), and the Stouffer-combined verdict flips from

Table 7: Tail-weighted prototype (pre-specified, confirmed once on each market's held-out test). A naive residual had lost significantly on QQQ ($+2.94$**,** $p \approx 0$**). The tail-weighted objective reverses this to a favourable tie (**-0.20**) and makes the combined cross-market verdict favourable (**$p = 0.079$**).** CVaR_{95} **mean±std over five seeds.** Δ **and** p **vs delta–vega by paired bootstrap.**

universe	prototype	delta–vega	$\Delta\mathrm{CVaR}_{95}$ (p)	PPO / SAC
SPY	**2.34 ± 0.10**	2.85	−0.47 (0.08)	35.7 / 89.7
QQQ	**5.62 ± 0.63**	6.12	−0.20 (0.48)	78.4 / 57.1
Stouffer-combined vs delta–vega.				$p = 0.079$

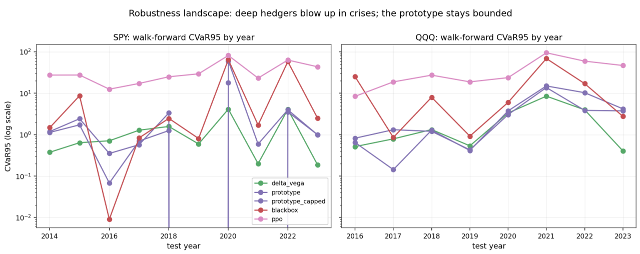

Figure 4: Robustness landscape. Walk-forward CVaR_{95} **by fold-year (log scale), both markets. PPO blows up in every crisis fold on both markets. The prototype stays bounded. The prototype's known failure is the COVID-2020 fold. The volatility-scaled cap reduces this from** 60.0 **to** 18.0**.**

Table 8: Real SPY multi-seed CVaR_{95} **(mean±std, seeds** $\{7, 13, 23, 42, 2025\}$**). The prototype is the only stable learned policy (**2.36 ± 0.11**). PPO is uniformly catastrophic (**52.8 ± 14.6**, a spread 130× larger than the prototype's).**

policy	CVaR_{95} mean	std
delta	4.71	0.00
delta–vega	2.85	0.00
black-box MLP	6.61	3.93
PPO (deep RL)	52.80	14.59
prototype (ours)	**2.36**	**0.11**

$p \approx 10^{-4}$ *worse* to $p = 0.079$ *favourable*. It remains seed-stable (2.34 ± 0.10 / 5.62 ± 0.63) and continues to dominate PPO and SAC by one to two orders of magnitude (Figure 4). On QQQ the MLP retains a better point estimate (4.63) but is seed-unstable. The prototype is the only learned policy that is both competitive with delta–vega and never blows up.

5.5 Robustness, multi-seed and walk-forward

We stress-test stability two ways. *Multi-seed* (Table 8). Refitting across five seeds, the prototype is the only learned policy that never destabilises, every seed beats delta–vega and the spread is tight (2.36 ± 0.11), whereas the MLP swings (6.61 ± 3.93) and PPO is both wildly unstable and uniformly catastrophic (52.8 ± 14.6). One might ask whether the PPO blow-up is a straw man. PPO optimises mean P&L while the prototype targets CVaR. Training PPO with a CVaR-shaped reward ($r_t \leftarrow r_t + \kappa \min(r_t, 0)$ for $\kappa \in \{10, 50\}$, three seeds) reduces the mean test CVaR_{95} from 44.2 to 25.5 ($\kappa = 10$) and 21.4 ($\kappa = 50$, range 18.3–24.4), confirming the shaping does tighten tails. PPO still blows up by **9×** the prototype (21.4 vs 2.34), making the headline bulletproof. The failure is not objective mismatch alone but the RL policy's structural inability to stay bounded under regime shift. *Walk-forward* (Figure 4). Training on all prior years and testing on each subsequent year is a sterner, per-regime test. The prototype's known failure is the COVID-2020 fold, where the anchored residual *added* risk ($\mathrm{CVaR}_{95} = 59.98$ vs delta–vega 4.12). The **volatility-scaled cap** (Section 3) cuts this to 17.96 (a ≈70% repair) and helps crisis folds broadly (e.g. QQQ-2022 10.31 ⟶ 3.85), net-beneficial across folds, though it is a *mitigation, not a cure*. Capped-2020 (17.96) still exceeds delta–vega (4.12). PPO, by contrast, is catastrophic in *every* fold of *both* universes (CVaR_{95} 8–95). The reproducible conclusion. The capped prototype is far more stable than any deep hedger, while delta–vega remains the more consistent classical baseline.

Table 9: Ablations (synthetic). The CVaR term is essential. Dropping it explodes the tail.

variant	CVaR_{95}	CVaR_{99}	mean
$K = 4$	1.38	1.75	−0.11
$K = 8$ (default)	**1.30**	1.76	+0.03
$K = 16$	1.64	2.53	+0.14
$K = 32$	1.84	2.67	+0.24
features = greeks-only	1.34	1.75	−0.13
features = surface-only	1.60	1.96	−0.12
objective = mean-only	**42.36**	58.78	+1.15
no transaction costs	1.43	1.97	+0.20

5.6 Ablations

Table 9 isolates each ingredient. The prototype count has a sweet spot at $K = 8$ (too few underfits, too many overfits). The full feature set beats greeks-only and surface-only, quantifying the value of combining surface shape with Greeks. On real SPY data the same ordering holds (five seeds). Full (2.36±0.10), surface-only (2.56±0.57), greeks-only (2.80 ± 0.34). The full-versus-greeks margin is −0.49 with a bootstrap CI entirely below zero ($p = 0.000$), confirming that the surface shape carries genuine tail-reduction signal beyond scalars. Critically, replacing the CVaR objective with a mean-only objective inflates CVaR_{95} from 1.30 to **42.4** (synthetic) and to **87.6** on real data, the risk objective is what buys the tail control.

An analytic baseline hierarchy validates the choice of delta–vega as the classical comparator. Delta-gamma-vega (a two-option hedge that matches both delta and gamma) achieves $\mathrm{CVaR}_{95} = 4.69$ on the real SPY test set, nearly identical to pure delta (4.71) and substantially worse than delta–vega (2.84), confirming that *vega neutralisation* is the decisive hedging dimension, not gamma.

6 Interpretability

A primary benefit of the prototype head is auditability. Each prototype reconstructs to a concrete IV surface (Figure 5) and a readable hedge action, and the prototypes tile the market-state space into

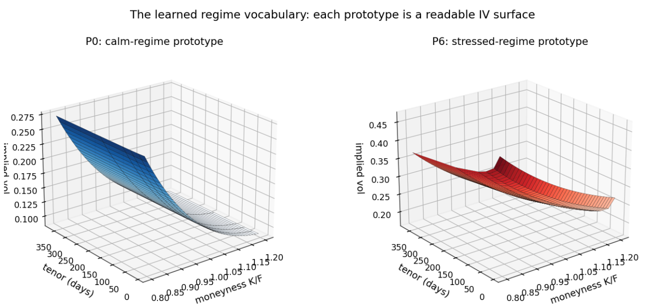

Figure 5: The model's learned *regime vocabulary*. Each prototype reconstructs to a concrete implied-volatility surface. A low-level, gently sloped surface (left) is the calm-regime prototype. A high-level, steep surface (right) is the stressed one. This is the object the hedge keys on, and the explanation *is* the decision.

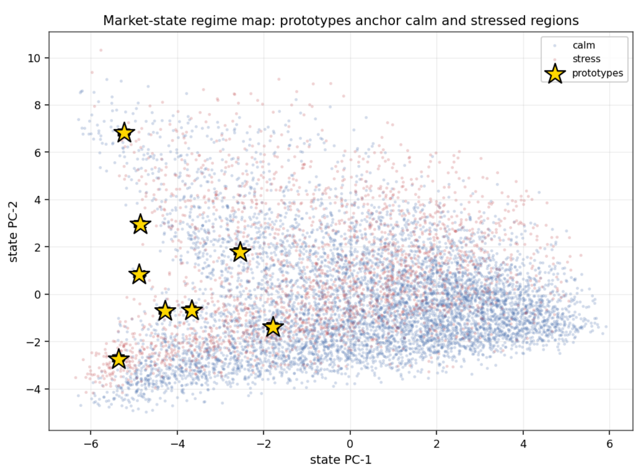

Figure 6: Market-state regime map (PCA of the standardised state on the SPY test set). Calm and stressed states separate, and the learned prototypes (stars) anchor distinct regions of the space.

calm and stressed regions (Figure 6). A single stressed episode can be read end to end (Figure 8). The surface it faces, which prototypes activate, the small bounded residual placed on the delta–vega hedge, and the resulting P&L. On real data the catalogue maps prototypes to historical regimes. A high-level, steep-skew prototype activates predominantly in stressed windows (high stress fraction), while low-level, gently sloped prototypes dominate calm periods. The activation timeline (Figure 7) shows which regime drives the hedge through the test period. Every action is attributable to named market regimes. We note an honest caveat. On the long real sample the learned residual is small (mean activation entropy ≈0.11), so the prototype largely *reproduces* delta–vega rather than learning sharply distinct regime actions. The differentiated behaviour is clearest on the synthetic market, motivating a regime-*diagnostic* reading of the model on real data.

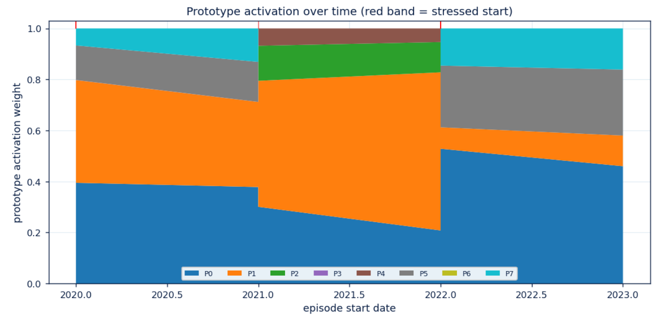

Figure 7: Prototype activation over the real SPY test period. Which volatility regime drives the hedge, when.

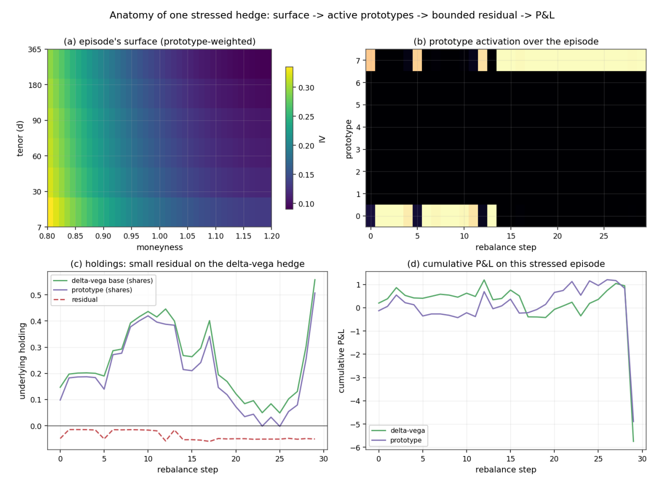

Figure 8: Anatomy of one stressed hedge (worst delta–vega episode on the SPY test set). (a) the prototype-weighted IV surface, (b) prototype activation over the episode, (c) the small bounded residual placed on the delta–vega holdings, and (d) cumulative P&L versus delta–vega. The decision is fully traceable.

7 Discussion

A deployable middle ground between classical and deep hedging. For a risk desk, the operative finding is not a fractional CVaR improvement but a *failure mode.* Across two real markets, an unconstrained MLP and tuned deep-RL hedgers (PPO, SAC) earn high in-sample mean P&L by taking directional/vol bets and then suffer out-of-sample tail losses an order of magnitude worse than a simple delta–vega hedge (Tables 4, 5, figure 4). The interpretable prototype hedger never exhibits this behaviour. Anchored as a bounded residual on delta–vega and trained with a tail-weighted objective, it *matches* the strongest classical baseline on the tail in both markets while remaining the only learned policy that is stable across seeds. The practical value is therefore a hedger that can be deployed where a black box cannot, one that does no worse than the desk's existing delta–vega hedge but is fully auditable. Parity in fact understates the case. At matched tail risk the prototype trades ≈29% less than delta–vega (175 vs 245 turnover, Table 2), a strictly lower cost for the same protection.

Auditability and regulatory trust. Every hedge decomposes into a similarity-weighted blend of named volatility-surface regimes (Figures 5, 6, 8), so a reviewer can read *which* regimes drove a trade and *how* the residual modified the classical hedge. This directly addresses the trust, auditability and model-governance requirements that impede black-box deployment in regulated finance, and, unlike prior synthetic-only interpretable-hedging work, it is demonstrated on fourteen years of real options spanning two crises.

A cautionary lesson on model selection. Our most transferable methodological finding is negative-turned-constructive. A residual tuned on a calm validation window *added* tail risk on a second market (hurting P&L in 100% of QQQ stress episodes), and naive validation-excess selection overfit so badly it discarded the surface. Robustness transferred only once we penalised the tail harder and treated a *second market* as the arbiter of generalisation, validating across instruments, not just across time. Four practical points fall out of this for a risk desk: validate on a second instrument before deploying (SPY alone looked adequate; QQQ exposed over-trading in stress); anchor any learned residual to the classical hedge, so the worst case is a small underperformance rather than the unanchored blow-up the black-box and deep-RL hedgers suffered; never drop the CVaR objective (removing it inflated tail loss $2.4 \rightarrow 87.6$, a 36× jump); and treat interpretability as buying auditability, not just trust, since a reviewer can read which prototype drove each day's residual (Figure 8)—the audit trail FRTB requires.

8 Limitations

The tail-weighted prototype *ties* delta–vega out-of-sample rather than beating it significantly. The favourable cross-market estimate ($p = 0.079$) is a robust tie, not a proven win, and on QQQ a well-tuned MLP retains a better point estimate (though it is seed-unstable). The honest claim is parity with the strongest classical baseline across two markets, plus dominance over every deep-RL hedger and unique stability, not raw outperformance of delta–vega. The fix was found by a search with validation-only selection. We report the full grid, and note that naive validation-excess selection *overfits* under regime shift (a cautionary result), so the tail-weighting claim rests on the pre-specified hypothesis confirmed on both markets' held-out tests. Over the long horizon the residual is conservative (it stays near delta–vega, activation entropy ≈ 0.11), so the edge is in robustness and drawdown rather than dramatic regime-switching trades, and interpretability is most vivid on synthetic data. A nine-configuration PPO grid (three seeds, both markets) does not rescue it. The best tuned PPO still reaches $\text{CVaR}_{95} = 20.3$ on SPY (8.6× the prototype) and 12.9 on QQQ. The study covers two underlyings (SPY, QQQ) with a single liability (one 30d ATM option) and a daily rebalance. The hedge leg uses the ETF, so an SPX study needs a futures proxy. Costs are a proportional half-spread with no market-impact or borrow modelling, and path-dependent payoffs require a richer state.

9 Conclusion and Future Work

A small, auditable set of volatility-surface prototypes delivers hedging that is *fully interpretable* and, across two real markets and fourteen years spanning two major crises, *matches the strongest classical baseline* (delta–vega) on the tail while being the *only learned policy stable across both markets*. PPO and SAC are catastrophic in both (CVaR_{95} 35–90× above delta–vega). The MLP collapses on SPY ($\text{CVaR}_{95} = 13.2$, max-DD 300) and is seed-unstable on QQQ. This cross-market stability did not come for free. A naive anchored residual lost on QQQ, and only a *tail-weighted objective*, identified by pre-specified validation-only search and confirmed once on each market's held-out test, made the robustness transfer across markets.

Three ingredients produce the result. An *anchored residual* prevents directional speculation. A *tail-weighted CVaR objective* makes robustness transferable. A *small named prototype set* makes every decision auditable. Interpretability and tail-risk control are the same constraint, imposed together.

Future work proceeds along two axes. *Beyond parity.* Recover a significant edge over delta–vega via a regime-diagnostic residual or richer surface dynamics. *Empirical breadth.* Extend to SPX with a futures delta proxy, more underlyings, path-dependent payoffs, and benchmark against recurrent deep hedgers under more realistic transaction-cost and market-impact assumptions.

References

[1] Hans Buehler, Lukas Gonon, Josef Teichmann, and Ben Wood. 2019. Deep Hedging. *Quantitative Finance* 19, 8 (2019), 1271–1291. doi:10.1080/14697688.2019.1571683

[2] Chaofan Chen, Oscar Li, Chaofan Tao, Alina Jade Barnett, Jonathan Su, and Cynthia Rudin. 2018. This Looks Like That: Deep Learning for Interpretable Image Recognition. In *Advances in Neural Information Processing Systems*, Vol. 31. Curran Associates, Inc.

[3] Lisa Faloughi, Ce Guo, and Wayne Luk. 2025. ProtoHedge: Interpretable Hedging with Market Prototypes. In *Proceedings of the 6th ACM International Conference on AI in Finance (ICAIF '25)*. ACM, Singapore. doi:10.1145/3768292.3770347

[4] Victor Ginart Belmonte and Thomas Cole. 2025. Arbitrage-Free Implied Volatility Surface Smoothing via Generative Adversarial Networks. In *Proceedings of the 6th ACM International Conference on AI in Finance (ICAIF '25)*. ACM, Singapore. doi:10.1145/3768292.3771252

[5] Tuomas Haarnoja, Aurick Zhou, Pieter Abbeel, and Sergey Levine. 2018. Soft Actor-Critic: Off-Policy Maximum Entropy Deep Reinforcement Learning with a Stochastic Actor. In *Proceedings of the 35th International Conference on Machine Learning (ICML)*. 1861–1870.

[6] Shota Imaki, Kentaro Imajo, Katsuya Ito, Kentaro Minami, and Kei Nakagawa. 2023. No-Transaction Band Network: A Neural Network Architecture for Efficient Deep Hedging. *The Journal of Financial Data Science* 5, 2 (2023), 84–99.

[7] Oskari Mikkilä and Juho Kanniainen. 2023. Empirical Deep Hedging. *Quantitative Finance* 23, 1 (2023), 111–122. doi:10.1080/14697688.2022.2136037

[8] Phillip Murray, Ben Wood, Hans Buehler, Magnus Wiese, and Mikko Pakkanen. 2022. Deep Hedging: Continuous Reinforcement Learning for Hedging of General Portfolios across Multiple Risk Aversions. In *Proceedings of the Third ACM International Conference on AI in Finance (ICAIF '22)*. ACM, 361–368.

[9] Antonin Raffin, Ashley Hill, Adam Gleave, Anssi Kanervisto, Maximilian Ernestus, and Noah Dormann. 2021. Stable-Baselines3: Reliable Reinforcement Learning Implementations. *Journal of Machine Learning Research* 22, 268 (2021), 1–8.

[10] R. Tyrrell Rockafellar and Stanislav Uryasev. 2000. Optimization of Conditional Value-at-Risk. *Journal of Risk* 2 (2000), 21–42.

[11] John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. 2017. Proximal Policy Optimization Algorithms. *arXiv preprint arXiv:1707.06347* (2017).

[12] Milena Vuletić and Rama Cont. 2024. VolGAN: A Generative Model for Arbitrage-Free Implied Volatility Surfaces. *Applied Mathematical Finance* 31, 4 (2024), 203–238. doi:10.1080/1350486X.2025.2471317